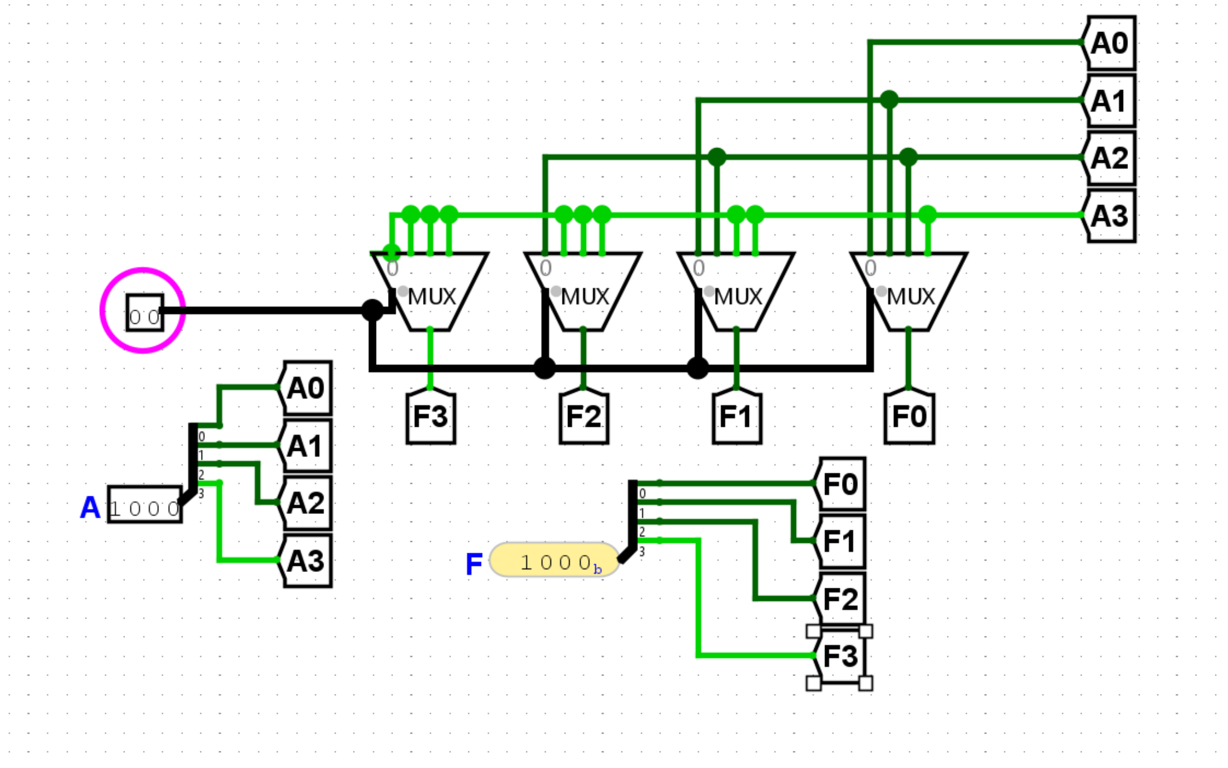
Roll-2103064

Set-B

1. **Arithmetic Right Shift:**



2. **4-bit parallel full Subtractor:**

#module file:

module sub1 (
  input wire A,
  input wire B,
  input wire Bin,
  output reg D,
  output reg Bout
);
	module Full_sub (
	  input wire [3:0]A,B,
	  input wire Bin,
	  output wire [3:0]D,Bout
	);
	
	sub1 s1(A[0],B[0],Bin,D[0],Bout[0]);
	sub1 s1(A[1],B[1],Bout[0],D[1],Bout[1])

assign D=A^B^Bin;
assign Bout=Bin&(~(A^B))+(~A)&B;
 	
endmodule
	sub1 s1(A[2],B[2],Bout[1],D[2],Bout[2])
	sub1 s1(A[3],B[3],Bout[2],D[3],Bout[3])
	
	endmodule

Test_bench file:

```
`timescale 1ns/1ps
module fs_tb;
reg [3:0]A;
reg [3:0]B;
reg Bin;
wire [3:0]Bout;
wire [3:0]D;

Full_sub uut(A,B,Bin,D,Bout);

initial begin
  $dumpfile("test.vcd");
  $dumpvars(0,fs_tb);
  A=4'b1000;
  B=4'b0011;
  Bin=0;
  #20;

  #40;
end

initial begin

$monitor("A=%b, B=%b Bin=%b D=%b Bout=%b",A,B,Bin,D,Bout);
  end
```

```
endmodule
```